

CS & IT ENGINEERING

Algorithms

Analysis of Algorithms

DPP 01

(Discussion Notes)



By- Aditya sir



[MSQ] → multiple

#Q. Which of the following notation is/are transitive but not reflexive

A Big oh (O) ✗

B Big omega (Ω) ✗

C Small oh (o) ✓

D Small omega (ω) ✓

Transitive

Reflexive

✓

✗

Ans: C, D

	0	Ω	$\emptyset$	0	ω
Reflexive	✓	✓	✓	×	×
Transitive	✓	✓	✓	✓	✓

[MSQ] $\rightarrow$ multiple

#Q.

$$\text{If } f(n) = \sum_{i=1}^n i^3$$

Ans:- A, B, D

Then which of the following choices is/are true for $f(n)$?

☒ **A**

$$\theta(n^4)$$

$O(n^4) \rightarrow$ Tight UB

$\Omega(n^4) \rightarrow$ Tight LB

☒ **B**

$$\Omega(n^4)$$

$$= O(n^4) \checkmark$$

$$= O(n^5) \checkmark$$

☒ **C**

$$\theta(n^5)$$

$$O(n^5) \checkmark$$

$$\Omega(n^5) \times$$

☒ **D**

$$\Omega(n^3)$$

$$= O(n^6) \checkmark$$

$$\Omega(n^5) \times$$

$$\Omega(n^4) \times$$

$$\Omega(n^3) \checkmark$$

$$\Omega(n^2) \times$$

$\vdots$

Soln :-

$$f(n) = \sum_{i=1}^n i^3$$

$$= 1^3 + 2^3 + 3^3 + \dots + n^3$$

$$= \left[\frac{n(n+1)}{2} \right]^2$$

$$= \left(\frac{n^2 + n}{2} \right)^2$$

$$= n^4 + \dots$$

$$= \underline{\underline{\Theta(n^4)}}$$

$$(a+b)^2 = \underline{\underline{a^2 + 2ab + b^2}}$$

[MCQ]

#Q. Consider the following program:

```
main ( )
{
    P = n!
    for (i = 1; i ≤ n; ++i)
        for (j = 1; j ≤ P; ++j) j = j * 2
            C = C + 1;
}
```

log₂P

What is the time complexity of above code?

A $O(n^2)$ ✗

C $O(n \log n)$ ✗

B $O(n^2 \log n)$

D $O(n)$ ✗

Ans :- B

Soln:

for ($i=1; i \leq n; i++$) $\longrightarrow O(n) \rightarrow$ runs n times

{
| for ($j=1; j \leq P; j=j*2$) $\longrightarrow$ runs
| $\log_2 P$ times
}

3

Overall TC $\Rightarrow n * \log_2 P$

given $P = n!$ $\approx \underline{\underline{n^n}}$

$n * \log_2 n^n$

$n * n * \log n$

$= O(n^2 \log n)$

[MCQ]

#Q. Consider the following code:

```
main ( )
{
    i = 1; j = 1
    while (j ≤ n )
    {
        ++ i;
        j = j + 1;
    }
```

$O(k)$
 $\Rightarrow \underline{O(\sqrt{n})}$ ✓

What is the time complexity of above code?

Ans :- B

A

$\theta(n)$

B

$\theta(\sqrt{n})$

C

$\theta(\log)$

D

$\theta(n \log(\log n))$

Soln: $\bar{i}=1, \hat{j}=1$

while ($\hat{j} \leq n$)
 {
 $\bar{i}++$
 $\hat{j} = \hat{j} + \bar{i}$
 }

$\bar{i}=1 \rightarrow \bar{i}=2 \rightarrow \bar{i}=3$
 $\hat{j}=1 \quad \hat{j} = \underline{1+2} \rightarrow \hat{j} = (1+2) + 3$

$\bar{i}=4 \quad \dots$
 $\hat{j} = 1 \rightarrow 4$

let us assume that loop runs
 for k times.

$\bar{i}=k$
 $\downarrow$

$\hat{j} = 1+2+3 \dots + k \implies n$

$\sum_{\bar{i}=1}^k \bar{i} = n$

$\boxed{\frac{k(k+1)}{2}} = n$

$\approx$

$k^2 = n$

$\boxed{k = \sqrt{n}}$



[MCQ]

#Q.

Consider the following code:

Algorithm $T(n)$ $\rightarrow TC(n)$

{

✓ if ($n = 1$) return; $\rightarrow O(1) \rightarrow \underline{B.C}$
else

{

$T(n/2)$; $\rightarrow TC(n/2)$
}

}

Ans: A

What is the space complexity of above code?

A

$\theta(\log n)$

B

$\theta(n)$

C

$\theta(n \log \log n)$

D

$\theta(\sqrt{n})$

Soln :- Using Back - Substitution logic.

Step 1 :- Algo $\rightarrow$ Time Complexity Recurrence

$$T(n) = a, \quad n=1$$

$$T(n) = T(n/2) + b, \quad n > 1$$

Step 2 :- Solve Recurrence using Back Substitution.

$$\begin{array}{l} T(n) = T(n/2) + b \quad \rightarrow (1) \\ \quad \quad \quad \downarrow \\ T(n/2) = T(n/2^2) + b \end{array}$$

$$T(n) = T(n/2^2) + b + b$$

$$\begin{aligned} T(n) &= T(n/2^2) + 2b \\ T(n/2^2) &= T(n/2^3) + b \end{aligned}$$

$$T(n) = T(n/2^3) + 3b \longrightarrow (2)$$

$$\vdots$$

General Term $T(n) = T(n/2^k) + k * b$

for Base condition, $n/2^k = 1$

$$n = 2^k \longrightarrow k = \log_2 n$$

$$\begin{aligned} T(n) &= T(1) + k * b \\ &= a + \log_2 n * b \end{aligned}$$

$$T(n) \longrightarrow \boxed{O(\log_2 n)}$$

[MSQ] $\rightarrow$ multiple

#Q. $f(n) = 2^{n^2}$, $g(n) = n!$ ^h~~h~~ $(n) = 2^{\log n^2}$

Ans: A, D

Which of the following is/are correct?

A $f(n) = \Omega(g(n))$ ✓

B $h(n) = \Omega(g(n))$ ✗

C $h(n) = O(g(n))$ ✓

D $g(n) = \Omega(f(n))$ ✗

Asymptotic Comparisons

$h(n) = O(g(n))$

$h(n) \leq c * g(n)$ ✓ $\rightarrow$ True.

$g(n) = \Omega(f(n))$

$g(n) \geq f(n) \rightarrow$ False

Soln:- $f(n) = 2^{n^2}$

$$g(n) = n! \approx n^n$$

$$h(n) = 2^{\log n^2} = 2^{2 \log n}$$

$2^{n^2} > n^n$

2^{n^2} vs n^n

$\log(2^{n^2})$ $\log(n^n)$

n^2 $n \log n$
 ~~$n \times n$~~ ~~$n \times \log n$~~

$n > \log n$

$2^{2 \log n} < n^n$

Take log

$\log(2^{2 \log n})$ vs $\log(n^n)$

~~$2 \log n$~~ ~~$n \log n$~~

$2 < n$

Conclusion :-

$$2^{n^2} > n! > 2^{\log n^2}$$

$$\boxed{f(n) > \underline{g(n)} > h(n)}$$

Check option A) :- $f(n) = \Omega(g(n))$

$$f \gg c * g \rightarrow \text{True}$$

Check option B) :- $h(n) = \Omega(g(n))$

$$h(n) \gg c * g \rightarrow \underline{\text{False}}$$

[NAT]

→ Numerical

#Q.

Consider the following notations:

1. $\sqrt{\log n} = O(\log \log n)$ ✗
2. $\log n = \Omega\left(\frac{1}{n}\right)$ ✓
3. $n^2 = \theta(2^{2 \log n})$ ✓
4. $(0.061)^n = \theta(1.02)^n$ ✗

2 & 3 are correct

Ans: 2

How many notations is/are correct? 2.

check 1) $\sqrt{\log n} = O(\log(\log n))$

$$(\log n)^{1/2} \leq c * \log(\log n)$$

let $\log n = x$

$\sqrt{x} \leq c * \log(x)$ X

False

$\sqrt{x} > \log x$

check 2 $\log n = \Omega(1/n)$

$$\log n \geq c * \frac{1}{n}$$

log ↑

Decre

$$\frac{1}{n} \rightarrow \left(\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5} \right)$$

Check 3) $n^2 = \Theta(2^{2\log n})$ ✓ equal rate of growth

$$n^2 \quad \text{vs} \quad 2^{2\log n}$$

Taking $\log_2$ both sides

$$\log_2(n^2) \qquad \log_2(2^{2\log n})$$

$$2\log n \qquad 2\log n * \log_2 2$$

$$\boxed{2 * \log n = 2 * \log n}$$

check 4)

$$\frac{(0.061)^n}{c_1^n} = \frac{0(1.02)^n}{c_2^n} \quad \times \rightarrow > 1$$

$$0.061 < 1$$

$$\frac{1}{2} = 0.5$$

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} = 0.25$$

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8} = \underline{\underline{0.125}}$$

Decr function

↓
Expo Decr

$$\underline{\underline{1.02 > 1}}$$

$$(>1)^n \rightarrow \underline{\text{Incr}}$$

$$\underline{\underline{2^2 < 2^3 < 2^4}}$$

↓
Expo incr.

[MCQ]

#Q. Consider the following functions:

$$f_1 = 2^n$$

$$f_2 = n!$$

$$f_3 = n^n$$

$$f_4 = e^n$$

Asymptotic
Comparison

What is the correct increasing order of above function?

A $f_1 f_4 f_2 f_3$ ✓

C $f_2 f_4 f_1 f_3$ ✗

B $f_2 f_1 f_4 f_3$ ✗

D $f_2 f_1 f_4 f_3$ ✗
 $f_2 f_1 f_3 f_4$

Ans. ∴ A

Soln:-

$$f_1 = 2^n \rightarrow \text{Expo}$$

$$f_2 = n! \rightarrow \text{Expo}$$

$$f_3 = n^n \rightarrow \text{Expo}$$

$$f_4 = e^n \rightarrow \text{Expo}$$

$$n! \approx n^n$$

$$n! < n^n$$

$$n(n-1) \dots 1 < n \times n \times n \dots$$

$$2^n \text{ vs } e^n$$

$e \approx 2.71$
 $e > 2$

$$f_3 > f_2$$

$$\underline{\underline{2^n < e^n}}$$

$$f_3 > f_2 \quad \text{and} \quad f_4 > f_1$$

$$f_3 > f_2 > f_4 > f_1$$

$$\underline{\underline{f_1 < f_4 < f_2 < f_3}}$$

way 1

$$\begin{array}{ccc} n! & vs & e^n \\ \Rightarrow n! & & = e^n \\ n! & & \\ n! & & \\ n^n & > & e^n \end{array}$$

way 2

$$\begin{array}{ccc} n^n & & e^n \\ \text{Taking } \log_e() & & \\ \log_e(n^n) & & \log_e(e^n) \\ n * \log_e n & > & n \end{array}$$

[MCQ]



#Q. Sort the functions in ascending order of asymptotic (big-O) complexity.

increasing

$$f_1(n) = n, f_2(n) = 80, f_3(n) = n^{\log n}, f_4(n) = \log \log^2 n, f_5(n) = (\log n)^{\log n}$$

A $f_2(n), f_4(n), f_1(n), f_5(n), f_3(n)$

A

B $f_2(n), f_1(n), f_4(n), f_5(n), f_3(n)$ ✗

C $f_2(n), f_1(n), f_4(n), f_3(n), f_5(n)$ ✗

D $f_2(n), f_1(n), f_4(n), f_3(n), f_2(n)$ ✗

Soln:

Rate of growth:

decr < Const < Poly < Expo

$f_1(n) = n$ —————→ Poly

$f_2(n) = 80$ —————→ Const

$f_3(n) = n^{(\log n)}$ —————→ Expo —

$f_4(n) = \log((\log n)^2)$ —————→ Poly

$f_5(n) = \log n^{(\log n)}$ —————→ Expo —

f_2 < f_4 < f_1 < f_5 < f_3

n $\log((\log n)^2)$

$n > 2 \log(\log n)$

$$n^{\log n} > \log n^{\log n}$$

$$\cancel{\log n} * \log n$$

$$\cancel{\log n} * \log(\log n)$$

$$\text{let } \log n \rightarrow x$$

$$\frac{\log n}{x}$$

$$>$$

$$\frac{\log(\log n)}{\log(x)}$$

[MSQ] $\rightarrow$ multiple Soln (can) be correct



#Q. Consider two function $f(n) = 10n + 2\log n$ and $g(n) = 5n + 2(\log n)^2$, then which of the following is correct option?

☒ **A** $f(n) = \theta(g(n))$

$f \leq g$
✓

☒ **B**

$f(n) = O(g(n))$ $\xrightarrow{\text{Big oh}}$

☐ **C** $f(n) = \omega(g(n^2))$ $\xrightarrow{\text{small omega}}$ X

$f(n) > g(n^2) * c$

X ☐ **D**

None of the above

(A, B) ✓

Soln: $\underline{f(n)} = \underline{10 \times n + 2 \times \log(n)}$, $g(n) = \underline{5n + 2(\log n)^2}$

$$\boxed{f(n) = \Theta(n)}$$

O Ω

$$\boxed{g(n) = \Theta(n)}$$

O Ω

$$\left. \begin{array}{l} f = \Theta(n) \\ g = \Theta(n) \end{array} \right\} \left. \begin{array}{l} f = \Theta(g(n)) \\ \underline{g = \Theta(f(n))} \end{array} \right\}$$

$$g(n) = 5n + 2(\log n)^2$$

$$g(n^2) = 5n^2 + 2(\log(n^2))^2$$

$$= 5n^2 + 2(2\log(n))^2$$

$$= \underline{5n^2} + 8(\log n)^2 \longrightarrow \underline{\underline{O(n^2)}}$$

$$f(n) = O(n), \quad g(n^2) = O(n^2)$$

$$f(n) < \underline{\underline{g(n^2)}}$$

#Q. Consider two function $f(n) = \sqrt{n}$ and $g(n) = n \log n + n$ then $f(n) / g(n)$ is equivalent to how many of the following given below? ____.

A $O(n^{-1/2})$ ✓ *Small oh*

B $O(n^{-1/2})$ ✓ *Big oh*

C $\Omega(1/\log n)$ ✗

D $\theta(n^{-1/2})$ ✗

$$\frac{f(n)}{g(n)} < \frac{1}{\sqrt{n}}$$

Ans: (2)

Soln:-

$$f(n) = \sqrt{n}$$

$$g(n) = n \log n + n$$

$$\boxed{h(n)} = \frac{f(n)}{g(n)} = \frac{\sqrt{n}}{n \log n + n} = \frac{\sqrt{n}}{n(\log n + 1)} = \boxed{\frac{1}{\sqrt{n}(\log n + 1)}}$$

Check option A: $h(n) = \frac{1}{\sqrt{n}(\log n + 1)}$ is $O(n^{-1/2})$?

$$\frac{1}{\log n + 1} < 1$$

$h(n)$ is not
 $\Omega(n^{-1/2})$

↓
Hence not
 $O(n^{-1/2})$

$$\frac{1}{\sqrt{n}(\log n + 1)}$$

~~$$\frac{1}{\sqrt{n}(\log n + 1)}$$~~

$$\frac{1}{(\log n + 1)}$$

$$n^{-1/2} \rightarrow \frac{1}{\sqrt{n}}$$

~~$$\frac{1}{\sqrt{n}}$$~~

Hence A & B
both are True

$$< 1$$

check ($\frac{1}{\sqrt{n}(\log n + 1)}$) is $\Omega\left(\frac{1}{\log n}\right)$?

$$\frac{1}{\sqrt{n} \log n + \sqrt{n}} < \frac{1}{\log n}$$

$$h(n) = \Omega\left(\frac{1}{\log n}\right)$$

X

$$\sqrt{n} \log n + \sqrt{n} > \log n$$

#Q. Consider the following C-code

```
void foo (int n)
{
    int a = 1;
    if (n == 1)
        return;  $\rightarrow$  exit
    for (; a <= n; a++) for (a = 1; a <= n; a++)
    {
        printf("GATEWALLAH");
        break;
    }
}
```

A $O(1)$ **B** $O(n)$ **C** $O(\log n)$ **D** $O\sqrt{n}$

Ans :- **A**

What is the worst time complexity of above program?

#Q. Consider the following asymptotic functions :

$$f_1 = 2^n$$

$$f_2 = 1.001^n$$

$$f_3 = e^n$$

$$f_4 = n!$$

Ans $\rightarrow$ (D)

Which of the following is correct increasing order of above functions?

A f_3, f_4, f_1, f_2 ✗

B f_2, f_4, f_1, f_3 ✗

C f_3, f_2, f_1, f_4 ✗

D f_2, f_1, f_3, f_4 ✓

Soln :

$$f_1 = 2^n \longrightarrow \text{Expo}$$

$$f_2 = (1.001)^n \longrightarrow \text{Expo}$$

$$f_3 = e^n \longrightarrow \text{Expo}$$

$$f_4 = n! \longrightarrow \text{Expo } (n^n)$$

$$L_1 \underbrace{(n(n-1) \times (n-2) \dots 1)}_{\approx n^n}$$

$$e \rightarrow \approx 2.71$$

$$1.001 < 2 < 2.71(e)$$

$$(1.001)^n < 2^n < e^n < n^n$$

$$f_2 < f_1 < f_3 < f_4$$

MSQ $\rightarrow$ Multiple can be correct
options

#Q. Consider ~~two function~~ the following functions:

$$f_1(n) = 4^{2^n}$$

$$f_2(n) = n!$$

$$f_3(n) = 4^{e^n}$$

$$f_4(n) = n^{n^n}$$

Ans: (B, C, D)

$$\underline{f_2} < \underline{f_1} < \underline{f_3} < \underline{f_4}$$

$$f_1 = O(f_2)$$

$$f_1 \leq f_2 \quad \times$$

Which of the following is/are correct?

A $f_1(n) = O(f_2(n))$ $\times$

B $f_1(n) = O(f_4(n))$

$$f_1(n) = O(f_4(n))$$

$$f_1 = O(f_4)$$

$$\downarrow$$

$$f_1 \leq f_4$$

C $f_1(n) = O(f_3(n))$

$$f_1 \leq f_3$$

D $f_2(n) = O(f_3(n))$

$$f_2(n) = O(f_3(n))$$

$$f_2 \leq f_3$$

Soln :-

$$\checkmark f_1(n) = (4)^{2^n} \longrightarrow \text{expo}$$

$$\underline{f_2(n) = n!} \longrightarrow (n)^n \longrightarrow \text{expo}$$

$$\checkmark f_3(n) = (4)^{e^n} \longrightarrow \text{expo}$$

$$f_4(n) = (n)^{n^n} \longrightarrow \text{expo}$$

$$f_2 < f_1 < f_3 < f_4$$

$$(4)^{2^n} < (4)^{e^n}$$

$$2^n < e^n \quad e \approx 2.71$$

$$f_1 < f_3$$

$$(n)^n < (n)^{n^n}$$

$$f_2 < f_4$$

$$(n)^n < (4)^{e^n}$$

$$\log(n^n)$$

$$\log((4)^{e^n})$$

$$n + \log n$$

$$e^n \times \log(4)$$

$$\text{Poly}(n \log n)$$

$$< \text{expo}(e^n)$$

$$(n)^n < (4)^{2^n}$$

$$\log(n^n)$$

$$\log((4)^{2^n})$$

$$n \times \log n$$

$$2^n \times \log(4)$$

$$\underbrace{(n \log n)}_{\text{Poly}} < \underbrace{(2^n)}_{\text{Expo}}$$

[MCQ]



Ans: (C)

#Q. Consider two function $f_1(n) = n^{2^n}$ and $f_2(n) = n^{n^2}$ then which of the following is true.

A $f_1(n) = O(f_2(n))$ ~~✗~~ $f_1 \leq f_2$

~~✗~~ **B** $f_1(n) = \theta(f_2(n))$

C $f_1(n) = \omega(f_2(n))$ $f_1 > f_2$
 (small omega)

~~✗~~ **D** None of these

$f_1 > f_2$

$f_1 \approx f_2$
 (rate of growth)

$2^n \neq O(n^2)$

Soln:-

$$f_1(n) = n^{2^n} > f_2(n) = n^{n^2}$$

log on both sides

$$\log(n^{2^n})$$

$$2^n \times \cancel{\log n}$$

$$\text{expo} \rightarrow 2^n$$

$$\log(n^{n^2})$$

$$n^2 \times \cancel{\log n}$$

$$\text{poly} \rightarrow n^2$$

[MCQ]



#Q. $f(n) = \sum_{i=1}^n i^3 = x$, choices for x

$$x = \frac{1}{4} (n^4 + 2n^3 + n^2)$$

$$x \rightarrow \left. \begin{array}{l} O(n^4) \\ \Omega(n^4) \end{array} \right\} \theta(n^4)$$

A I, II, III ~~X~~

(D)

B ~~X~~ II, III, IV

C ~~X~~ I, II, III, IV

D ~~X~~ I, III, IV

$$x = \Omega(n^3)$$

$$\checkmark x \geq c * n^3$$

$$x \leq n^4 \rightarrow \text{Tight}$$

$$x \leq n^5$$

$$\leq n^6 \rightarrow \text{loose}$$

$$x \rightarrow O(n^3) \checkmark$$

$$x \geq n^5 \times$$

Soln:

$$x = \sum_{i=1}^n i^3 = 1^3 + 2^3 + 3^3 \dots n^3$$

$$\rightarrow = \left(\frac{n(n+1)}{2} \right)^2 = \left(\frac{n^2+n}{2} \right)^2 \quad (n^2)^2 = n^{2 \times 2} = \underline{\underline{n^4}}$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$= \frac{1}{4} (n^2+n)^2 = \frac{1}{4} \left[(n^2)^2 + 2 \times n^2 \times n + n^2 \right]$$

$$= \frac{1}{4} \left[n^4 + 2n^3 + n^2 \right]$$

Dominating $\swarrow$

$\mathcal{O}(n^4)$



THANK - YOU

Practice all the
Concepts
taught in
Class